

Brijesh Yadav

Backend Software Engineer | Python • FastAPI • Distributed Systems • Trading Infrastructure

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Portfolio | GitHub | LinkedIn

Professional Summary

Backend Software Engineer with 4+ years of experience building scalable microservices, distributed systems, and real-time applications. Experienced in Java (Spring Boot), Python (FastAPI), PostgreSQL, Redis, WebSockets, REST APIs, and cloud-native architectures. Creator of an algorithmic trading platform and open-source trading SDKs used for broker authentication, market-data processing, and automated trading workflows. Strong foundation in system design, data structures, algorithms, concurrency, and high-performance backend engineering.

Technical Skills

Languages: Java, Python, SQL, JavaScript

Backend: Spring Boot, FastAPI, REST APIs, Microservices, Hibernate, JPA

Databases: PostgreSQL, MySQL, MongoDB, Redis

Distributed Systems: Kafka, Event-Driven Architecture, WebSockets, Caching

Cloud & DevOps: AWS S3, Docker, Git, Maven, CI/CD

Frontend: React

Core CS: Data Structures, Algorithms, System Design, OOP, Concurrency, Multithreading

Professional Experience

ArisGlobal

Oct 2021 – Present

Software Engineer

- Developed and maintained Spring Boot microservices supporting critical pharmacovigilance workflows used by global pharmaceutical organizations.
- Designed and implemented REST APIs and backend services enabling enterprise integrations and workflow automation.
- Built high-volume email ingestion and document-processing pipelines capable of processing thousands of inbound emails and attachments.
- Integrated OCR and NLP-based processing pipelines to transform unstructured business data into structured records.
- Improved processing efficiency by approximately 80% while achieving extraction accuracy above 95%.
- Optimized PostgreSQL and MySQL database queries, improving application performance and scalability.
- Investigated production issues, performed root-cause analysis, debugging, and delivered reliable fixes within Agile environments.

Projects

Growth Quantix – Algorithmic Trading Platform

Live Demo

GitHub Repository

Creator & Backend Engineer

- Architected and developed a multi-broker algorithmic trading platform supporting Upstox, Dhan, and Fyers integrations.
- Built backend services using FastAPI, PostgreSQL, Redis, and WebSockets for real-time market-data processing and strategy execution.
- Developed paper trading, backtesting, portfolio management, broker-account management, and strategy-execution modules.
- Designed scalable market-data ingestion pipelines for processing live stock-market feeds and distributing updates to connected users.
- Integrated Upstox Market Feed V3 WebSocket, historical candle APIs, broker authentication workflows, and automated trade execution services.
- Designed and implemented REST APIs supporting trading workflows, broker management, market-data access, portfolio analytics, and automated strategy execution.
- Built low-latency WebSocket infrastructure enabling real-time stock-price streaming and dashboard updates.

- Designed database schemas for users, orders, trades, strategies, broker configurations, trading sessions, AI predictions, and performance analytics.
- Applied asynchronous processing, Redis caching, and event-driven architecture patterns to improve throughput and responsiveness.
- Designed modular microservice architecture separating market data, broker integrations, trading execution, and user management services.

Technologies: Python, FastAPI, PostgreSQL, Redis, WebSockets, Docker, React, Broker APIs, JWT Authentication

Upstox Auth Pro – Open Source Python SDK

GitHub

PyPI Package

- Built and published an open-source Python SDK for Upstox OAuth2 authentication, achieving 2,600+ downloads on PyPI and simplifying automated broker integrations for trading applications.
- Implemented browser automation using Playwright with TOTP-based 2FA authentication and secure token generation.
- Designed pluggable token storage, automatic token refresh handling, retry mechanisms, and diagnostic logging.
- Built a production-ready CLI tool enabling automated authentication for algorithmic trading applications.
- Packaged and published the library to PyPI with automated CI/CD release workflows and comprehensive documentation.

Technologies: Python, Playwright, OAuth2, TOTP, PyPI, GitHub Actions, CLI Development

Upstox Auth Pro Java – Open Source Java SDK

GitHub

- Developed an open-source Java SDK for automating Upstox authentication and token management workflows.
- Implemented OAuth2 authentication, token refresh handling, and developer-friendly APIs for trading applications.
- Published documentation, examples, and integration guides for Java developers.
- Applied object-oriented design principles and modular architecture for maintainability and extensibility.

Technologies: Java, Maven, OAuth2, REST APIs

Prindox – Online Document Printing Platform

Live Demo

GitHub Repository

- Developed backend services and REST APIs supporting document upload, print-order management, and workflow automation.
- Designed PostgreSQL database schemas and optimized queries for efficient document storage and retrieval.
- Implemented asynchronous background-processing workflows for validation and order processing.
- Built scalable backend architecture focused on maintainability, extensibility, and operational efficiency.

Technologies: Python, FastAPI, PostgreSQL, REST APIs

Education

Rajkiya Engineering College Sonbhadra

2017 – 2020

Bachelor of Technology (Computer Science & Engineering)

CGPA: 7.5 / 10

Certifications

- Summer Internship – Data Driven Computing and Networking, MNNIT Allahabad